

Management of small bowel obstruction and therapeutic role of contrast media: A multicenter study.

Jaanimäe L, Lepner U, Kirsimägi Ü, Nikkolo C

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Background

Adhesive small bowel obstruction (SBO) accounts for 55–75% of mechanical small bowel obstructions [1,2]. According to guidelines, water-soluble contrast media (CM) is suggested in the diagnostic work-up of adhesive SBO [3], [4], [5]. However, there is currently no consensus on the therapeutic role of water-soluble CM in adhesive SBO [6] or a generally approved protocol as to when CM should be administered, how often should radiographs be taken and how long conservative management can safely be pursued.

Material and methods

The data of patients who were hospitalized for SBO in two regional hospitals of Estonia between January 2015 and December 2019 were retrospectively collected from the hospitals' electronic databases.

Results

Of 1,008 patients with mechanical SBO included in analysis, 140 (13.9%) were operated on immediately. In the case of conservative management, 575 (66.2%) patients received CM. Resolution occurred in 75.8% of the patients who received CM and in 68.9% of the patients who did not ($p = 0.037$). Resolution rate was the highest (78.8%) in the group that received CM within 12 h of admission. The duration of symptoms was longer in patients who needed surgical intervention compared to conservatively treated patients (31 h vs 15 h, $p < 0.001$). For patients with a history of a single previous abdominal/pelvic operation, the likelihood of surgical intervention was higher compared to patients whom resolution was achieved with conservative measures (50.9% vs 38.5%). In patients with three or more previous operations SBO resolved more often with conservative treatment compared to patients who required surgical intervention (22.7% vs 11.4%; $p < 0.001$).

Conclusion

The significantly higher rate of SBO resolution in the CM group suggests that CM may have a therapeutic role in SBO treatment, especially when it is administered within 12 h of admission. Patients with longer duration of symptoms and a history of one previous abdominal surgery need more likely surgical treatment.

Keywords

Introduction

Despite being an actual topic of numerous research papers, consensus on the role of water-soluble contrast media (CM) in management of SBO, as well as a unified protocol of how to guide its treatment are still lacking. Mechanical SBO is a condition that requires prompt evaluation and management. Adhesions are the most common cause of SBO [7]. Patient's primary evaluation should be conducted and the surgeon has to opt for a treatment strategy. Computed tomography (CT) is the recommended imaging technique, as it is more informative than plain radiograph [4]. Rebound tenderness and abdominal guarding are late signs [8] and, when associated with elevated C-reactive protein (CRP) >75 mg/L and white blood cell (WBC) count $>10 \times 10^9$ /L, alludes to abdominal emergency [9] and suggests the possible need for surgical exploration.

The Bologna guidelines suggest use of water-soluble CM as part of the management algorithm in the case of adhesive small bowel obstruction [4]. According to several studies, water-soluble CM has a therapeutic

effect due to its hyper-osmolar qualities that reduce bowel wall edema and help restore bowel motility [10], [11], [12], [13]. Koh et al., on the other hand, concludes that there remains a lack of evidence that water-soluble CM offers therapeutic benefit on its own. Despite the variation of conducted studies, there is no general consensus as to when water-soluble CM should be administered, nor are there recommendations about the timing of the decision to operate [6].

The main aim of the present multicenter retrospective study was to evaluate the association of CM on resolution of SBO. Another aim was to find predictive factors that could suggest the need for earlier operative management or predict the failure of conservative management.

Material and methods

The study was approved by the Research Ethics Committee of the University of Tartu.

This is a multicenter retrospective study conducted at two regional hospitals of Estonia: Tartu University Hospital and North Estonian Medical Center. The data of the patients' hospitalized for SBO between January 2015 and December 2019 were collected from electronic databases.

Eligible patients were adults aged 18 years or older with mechanical small bowel obstruction. Patients younger than 18 years, patients with large bowel obstruction or dynamic small bowel obstruction and/or patients who had undergone abdominal/pelvic surgery within 4 weeks of admission were excluded from the study. Patients diagnosed with incarcerated abdominal wall hernia were also excluded from analysis.

Based on the International Classification of Diseases 10th revision (ICD-10) codes K56.0-K56.7, patients with bowel obstruction were identified from electronic databases. All the case history and data were expatiated and recorded by the corresponding author and thus the inconsistent interpretation of data should be eliminated. The duration of symptoms (from symptom onset to the time of presentation to the emergency department) was documented. Characteristics of abdominal pain (absent/intermittent/permanent), presence of nausea/vomiting, absence of flatus, results of blood tests (white blood cell count (WBC), C-reactive protein (CRP), lactate level (Lac), estimated glomerular filtration rate (eGFR)), radiology imaging techniques used, previous abdominal/pelvic surgeries, previous episodes of bowel obstruction. It was documented whether immediate surgical treatment was implemented, or whether conservative management was initiated. In the case of conservative treatment, it was recorded whether contrast media (CM) was or was not used. In the case of surgical treatment, the type of surgery (adhesiolysis/ bowel resection/ other) was recorded.

The software package Statistica version 13.3 (TIBCO Software, California, USA) was used for statistical calculations. Categorical variables were given as count and percentages, comparison between the study groups was made using Pearson's Chi square test or Fischer's exact test. Continuous data were presented as means with standard deviation (SD) for normally distributed variables, or medians with 25% and 75% percentiles in the case of a non-normal distribution. When comparing two groups (operative vs conservative treatment) Student's t-test was used for the means and nonparametric Mann-Whitney U test was used for non-normally distributed variables. ANOVA and non-parametric Kruskal-Wallis tests were employed when three different conservative treatment study groups were compared. All statistical tests were two-sided and $p < 0.05$ was considered statistically significant.

Results

During a five-year period (2015–2019) 1115 patients with mechanical SBO were admitted to two regional hospitals of Estonia. The most common cause of mechanical SBO was adhesions (63.5%) (Table 1). Table 1 Etiology of mechanical small bowel obstruction. Table 1 Etiology of SBO n = 1115 (%) Adhesions 708 (63.5) Incarcerated abdominal wall hernia 107 (9.6) Carcinomatosis/tumor recurrence 106 (9.5) Unknown cause 105 (9.4) Internal hernia a 37 (3.3) Intestinal lumen stenosis 20 (1.8) Gallstone ileus 13 (1.2) Volvulus 9 (0.8) Meckel's diverticulum 7 (0.6) Invagination 2 (0.2) Diaphragmatic defect b 1 (0.1) SBO – small bowel obstruction. a 32 had a history of gastric bypass surgery, 4 had obturator hernias, 1 had undergone Roux-en-Y reconstruction. b Traumatic diaphragm rupture causing small bowel entrapment.

Etiology of mechanical small bowel obstruction.

SBO – small bowel obstruction.

32 had a history of gastric bypass surgery, 4 had obturator hernias, 1 had undergone Roux-en-Y reconstruction.

Traumatic diaphragm rupture causing small bowel entrapment.

A total of 107 patients were diagnosed with incarcerated abdominal wall hernia and were excluded from further analysis (Fig. 1). Fig. 1 flowchart of patients' management SBO – small bowel obstruction CM – contrast media given No CM – no contrast media given. Fig. 1

flowchart of patients' management

SBO – small bowel obstruction

CM – contrast media given

No CM – no contrast media given.

Plain radiograph was taken in 483 cases and CT (with or without intravenous contrast media) was done in 736 cases. A CT scan followed a previous radiograph in 228 cases.

Of the 1008 patients included in the analysis, 140 were operated within 6 h of admission. The remaining 868 were managed conservatively (nil per mouth, infusion therapy, analgetics) (Fig. 1). A nasogastric tube was inserted in 697 patients as part of conservative treatment, 159 were left without it and in 12 cases the data was missing.

In the group of conservative treatment 66.2% of patients were given CM (water-soluble CM in 561 cases, Barium in 14 cases). Altogether 391 patients received CM within 12 h of admission and 184 patients received it at a later time (from 13 to 120 h; median time 21 h). In 33.8% of the cases the contrast media challenge was not commenced (Fig. 1). There were no predetermined protocols as to how many radiographs are to be made and what should be the interval between them.

Out of the 293 patients who did not receive contrast media, SBO resolved in 68.9% of the cases. Of the patients who were given CM within 12 h of admission 78.8% had resolution. In the patient group who received CM later than 12 h of hospitalization, resolution occurred in 69.6% (Fig. 1). Overall, resolution occurred in 75.8% of the patients who received CM and in 68.9% of those who did not receive CM ($p = 0.037$).

When the three groups of conservative treatment (no CM, CM given within 12 h and CM given later than 12 h of admission) were compared, their symptoms and clinical finding at admission were similar (Table 2). The median time of onset of symptoms was the shortest in the group that did not receive CM. There was a significant difference in the values of WBC and CRP between the groups ($p < 0.05$). No difference in the previous occurrence of SBO was found. However, less patients had been operated for SBO previously in the no CM group compared to patients who received CM (16% vs 26% and 23%; $p = 0.017$) (Table 2). Table 2 Comparison of the study groups of conservative treatment. Table 2 Patients' data

	no CM (n = 293)	CM < 12 h (n = 391)	CM > 12 h (n = 184)	p-value
Sex: Female (%)	158 (53.9)	135 (46.1)	204 (52.2)	187 (47.8)
Male	100 (54.4)	84 (45.7)	0.850	
Mean age, years (SD)	65.3 (16.6)	64.4 (16.7)	66.4 (14.6)	0.377
Median time from onset of symptoms, hours (LQ; UQ)	15 (9; 48)	20 (8; 48)	24 (10; 70)	0.057
Previous SBO (%)	a130/266 (48.9)	172/336 (51.2)	69/154 (44.8)	0.421
Previously operated SBO (%)	a43/265 (16.2)	86/333 (25.8)	35/153 (22.9)	0.017
Previous operations (%)	aNo operations	1 operation	2 operations	≥ 3 operations
	31/270 (11.5)	121/270 (44.8)	71/270 (26.3)	47/270 (17.4)
	41/378 (10.9)	148/378 (39.1)	103/378 (27.3)	86/378 (22.7)
	14/174 (8.0)	78/174 (44.8)	49/174 (28.2)	33/174 (19.0)
Mean WBC, 10 ⁹ /L (SD)	b11.4 (7.1)	10.2 (3.8)	9.4 (4.6)	<0.001
Median CRP, mg/L (LQ; UQ)	b6 (2; 31)	5 (2; 22)	11 (2; 42)	0.011
Mean Lac, mmol/L (SD)	a,b1.84 (0.97)	(n = 70)	1.87 (1.08)	(n = 100)
Mean eGFR, mL/min/1.73 m ² (SD)	b75.1 (27.2)	78.6 (26.1)	74.7 (27.8)	0.143
Type of pain (%)	a, cIntermittent	Continuous	121/160 (76)	39/160 (24)
	161/230 (70.0)	69/230 (30.0)	64/102 (62.8)	38/102 (37.2)
Nausea/Vomiting (%)	a,c256/288 (88.9)	341/386 (88.3)	158/179 (88.3)	0.969
Absence of flatus (%)	a,c204/280 (72.9)	283/375 (75.5)	117/174 (67.2)	0.131
Clinical				

finding (%) a,cNo painRegional painDiffuse painGuarding/peritonitis61/272 (22.4)121/272 (44.5)86/272 (31.6)4/272 (1.5)69/371 (18.6)184/371 (49.6)114/371 (30.7)4/371 (1.1)36/168 (21.4)67/168 (39.9)65/168 (38.7)0/168 (0)0.204Median length of stay (LQ; UQ) d3 (2; 6)3 (2; 5)4 (3; 8)<0.001Died (%) e17 (5.8)11 (2.8)10 (5.4)0.122SD – standard deviation; LQ – lower quartile; UQ – upper quartile; SBO – small bowel obstruction; WBC – white blood cell count; CRP – C-reactive protein; Lac – lactate level; eGFR – estimated glomerular filtration rate.no CM – patients were not given contrast media.CM < 12 h – patients received contrast media within 12 h of admission.CM >12 h – patients received contrast media beyond 12 h of admission.aPatients whose data was missing were excluded from the analysis.bPrimary blood tests taken at admission.cSymptoms/clinical finding described at admission.dDays.eDuring hospitalization.

Comparison of the study groups of conservative treatment.

SD – standard deviation; LQ – lower quartile; UQ – upper quartile; SBO – small bowel obstruction; WBC – white blood cell count; CRP – C-reactive protein; Lac – lactate level; eGFR – estimated glomerular filtration rate.

no CM – patients were not given contrast media.

CM < 12 h – patients received contrast media within 12 h of admission.

CM >12 h – patients received contrast media beyond 12 h of admission.

Patients whose data was missing were excluded from the analysis.

Primary blood tests taken at admission.

Symptoms/clinical finding described at admission.

Days.

During hospitalization.

In summary of the 1008 patients with SBO, 641 (63.6%) patients were treated conservatively and 367 (36.4%) required surgery. The surgeries performed were categorized as adhesiolysis, bowel resection and other (detorsion of volvulus, bowel by-pass, stoma formation, enterotomy etc.). In the group of patients who were operated within 6 h of admission, adhesiolysis was performed in 23.6% of the cases, bowel resection in 32.8% of the cases and some other intervention was performed in 43.6% of the cases. After unsuccessful conservative management adhesiolysis was required in 36.6% of the cases, bowel resection was done in 26.9% and other interventions in 36.5% of the cases. The distribution of surgical interventions varies between the groups (p = 0.003).

Comparison of the groups of conservative and surgical treatment revealed there were no differences in the mean age or sex of the patients. The median time from the onset of symptoms (evaluated at admission) was significantly longer in patients who underwent surgical intervention (31 h vs 15 h, p<0.001). Nausea and inability to pass gas, i. e. common symptoms in SBO, occurred at a similar rate in both study groups. Continuous pain was more frequent in the group of surgical treatment (p<0.001). This correlates also with clinical finding at admission. Patients who presented with diffuse pain or symptoms of guarding/peritonitis were more likely to be operated than to be managed conservatively (p<0.001). There was a significant difference in the CRP and eGFR values between the two groups: median CRP was higher (12 mg/L vs 5 mg/L) and mean eGFR was lower (73.6 mL/min/1,73m² vs 78 mL/min/1,73m²) in the group of surgical treatment. Patients who had a history of SBO or had been operated for SBO previously were more likely to be successfully managed conservatively (p<0.001) (Table 3).Table 3Comparison of the groups of operative and conservative treatment.Table 3Patients' dataAll patients (n = 1008)Operative treatment group(n = 367)Conservative treatment group(n = 641)p-valueSex: Female (%)Male535 (53.1)473 (46.9)199 (54.2)168 (45.8)336 (52.4)305 (47.6)0.581Mean age, years (SD)64.7 (16.7)65.0 (16.3)64.6 (16.9)0.745Median time from onset of symptoms, hours (LQ; UQ)19 (9;48)31 (10;72)15 (8; 48)<0.001Previous SBO (%) a401/879 (45.6)96/310 (31.0)305/569 (53.6)<0.001Previously operated SBO (%) a182/873 (20.8)46/308 (14.9)136/565 (24.1)0.002Previous operations (%) aNo operations1 operation2 operations≥3 operations118/949 (12.4)408/949 (43.0)246/949 (25.9)177/949 (18.7)51/342 (14.9)174/342 (50.9)78/342 (22.8)39 /342

(11.4)67/607 (11.0)234/607 (38.6)168/607 (27.7)138/607 (22.7)<0.001Mean WBC, 109/L (SD) b10.5 (5.3)10.2 (4.6)10.7 (5.7)0.186Median CRP, mg/L (LQ; UQ) b6.3 (2; 29)12.0 (2; 57)5.0 (2; 20)<0.001Mean Lac, mmol/L (SD) a,b1.96 (1.2)(n = 70)2.00 (1.3)(n = 100)1.92 (1.1)(n = 37)0.621Mean eGFR, mL/min/1,73m² (SD) b76.4 (27.5)73.7 (29.3)78.0 (26.3)0.017Type of pain (%) a,cIntermittentContinuous389/582 (66.8)193/582 (33.2)126/215 (58.6)89/215 (41.4)263/367 (71.7)104/367 (28.3)0.001Nausea/vomiting (%) a,c874/985 (88.7)316/351 (90.0)558/634(88.0)0.338Absence of flatus (%) a,c676/953 (70.9)232/336 (69.1)444/617 (72.0)0.344Clinical finding (%) a,cNo painRegional painDiffuse painGuarding/peritonitis176/932 (18.9)418/932 (44.9)314/932 (33.7)24/932 (2.6)66/328 (20.1)112/328 (34.1)131/328 (40.0)19/328 (5.8)110/604 (18.2)306/604 (50.7)183/604 (30.3)5/604 (0.8)<0.001Median length of stay d (UQ; LQ)3 (2; 7)8 (5; 14)3 (2; 4)<0.001Died (%) d50 (4.9)29 (7.9)21 (3.2)0.002SD – standard deviation; LQ – lower quartile; UQ – upper quartile; SBO – small bowel obstruction; WBC – white blood cell count; CRP – C-reactive protein; Lac – lactate level; eGFR – estimated glomerular filtration rate.aPatients whose data was missing were excluded from the analysis.bPrimary blood tests taken at admission.cSymptoms/clinical finding described at admission.dDayseDuring hospitalization.

Comparison of the groups of operative and conservative treatment.

SD – standard deviation; LQ – lower quartile; UQ – upper quartile; SBO – small bowel obstruction; WBC – white blood cell count; CRP – C-reactive protein; Lac – lactate level; eGFR – estimated glomerular filtration rate.

Patients whose data was missing were excluded from the analysis.

Primary blood tests taken at admission.

Symptoms/clinical finding described at admission.

Days

eDuring hospitalization.

When comparing the history of earlier operations, the proportion of patients with no previous abdominal/pelvic surgeries was similar in both groups. Patients with a history of a single operation more often needed surgical intervention (50.9% vs 38.5%). The SBO in patients with three or more previous operations resolved more often with conservative treatment (22.7% vs 11.4%; $p<0.001$).

Discussion

Acute small bowel obstruction (SBO) accounts for about 20% of all surgical emergencies in Western societies [14]. During a five-year period, 1115 patients with mechanical SBO were treated at two regional hospitals of Estonia. According to literature data, the major cause of SBO is adhesions, the next two frequent causes being hernias and malignancies [15], which is also confirmed by the results of our retrospective study. However, in 105 cases, the cause of SBO remained unclear; 76 of those patients had no previous history of abdominal/pelvic surgery and in the remaining cases the data was missing. According to a recent review by Bayat et al. most SBOs in patients without previous abdominal surgery are adhesive in nature. The author suggested that future studies are needed to address the long-term outcomes and safety of nonoperative management for SBO in patients without previous abdominal surgery [16].

Plain radiograph (X-ray) was done 483 in cases out of 1008. According to the guidelines [4], computed tomography is a recommended diagnostic tool and it is also available in Estonian regional hospitals. CT scan (with or without intravenous contrast media) was done in 736 cases. Of the 483 patients in whom a plain radiograph was taken at admission, 228 underwent also a subsequent CT. This exposed the patients to unnecessary radiation. Thus it is preferable to use computed tomography, as it is more informative and can differentiate between the different causes of bowel obstruction [4]. The large number of plain radiographs and double investigations (X-ray + CT) might be caused by the fact that the study period began in 2015 when CT was not used as the primary tool of investigation. The updated Bologna guidelines were introduced in 2017 [4]. Since that time computed tomography has been the primary investigation performed at admission.

CM is used as part of the SBO management algorithm at regional hospitals of Estonia. However, the time of CM administration, as well as the number of radiographs and the interval between them vary greatly. In our study, the median time of CM administration was 8 h after hospitalization (minimum time 1 hour, maximum time 120 h). Resolution occurred in 75.8% of the patients who received CM and in 68.9% of the patients who did not receive CM ($p = 0.037$). According to our analysis, resolution occurred in 78.8% of the patients who received contrast media within 12 h of admission. The resolution rate was similar between the patient group that did not receive CM and the group that received it beyond 12 h of admission (68.9% and 69.6%, respectively). Considering this, it seems that CM administration shortly after admission could contribute to SBO resolution. In Zielinski's study, where 75% of patients passed the CM challenge, Gastrografin was independently associated with successful nonoperative management [11]. This supports the hypothesis that CM has an effect on SBO resolution. However, as the present study is retrospective in nature, we cannot make any definitive conclusions and further research is needed to establish the optimal time of CM administration.

Maximum time for SBO resolution in the present study was 96 h, which is in accordance with the findings by Mulder et al. according to which a significant proportion of SBOs require >48 h to resolve after CM [17].

Overall, of the 1008 patients who were hospitalized for SBO, surgery was performed in 367 cases. Predictive features suggesting that conservative treatment might prove unsuccessful are longer duration of symptoms, elevated CRP (>12 mg/L) and decreased glomerular filtration rate (<73.6 mL/min/1.73m²) at admission. Although there was a significant difference in the CRP and eGFR values between the study groups of surgical and conservative treatment, the difference was clinically not significant. Diffuse pain and peritonitis are alarming symptoms and suggest possible bowel strangulation which needs surgical management [4,3,18]. In our study 32.8% of the patients who were operated within 6 h of admission required bowel resection. This correlates with clinical finding and is an evidence of high surgical alertness – in the case of suspected bowel ischemia, operative treatment was implemented. Patients who had a history of previous SBO were more likely to be managed successfully by using conservative measures. This could be due to surgeons attempt to avoid complex surgery. Moran et al. stated that in patients with repeated SBO episodes and many prior laparotomies for adhesions, prolonged conservative treatment, including parenteral nutritional support may be prudent and often avoid a complex high-risk procedure [19].

According to Yang et al., in patients who were admitted several times for adhesive SBO, the relative risk of recurrent adhesive SBO increased with the increasing number of prior ASBO episodes [20]. Previous successful conservative management of SBO would allow the clinician to repeatedly use conservative measures, however with the odds of a higher recurrence rate [3].

The strength of this study is the representative study cohort. As the Estonian medical system is digitalized, the data was easily retrievable from electronic databases.

However, as this is a retrospective study, it has also specific limitations. Due to the slightly different management protocols between the hospitals, some data for the clinical objective evaluation of the patients was missing. Also, the timing of CM administration and the interval between the radiographs was variable. In 14 cases (2.4%) Barium was used instead of water-soluble CM. It is obvious that further randomized controlled studies are needed for obtaining more substantial evidence.

Conclusions

The significantly higher rate of SBO resolution in the CM group suggests that CM may have a therapeutic role in SBO treatment, especially when it is administered within 12 h of admission.

In patients with longer duration of SBO symptoms, particularly when they have a history of one previous abdominal/pelvic operation (possible single adhesion), conservative management is more likely to be unsuccessful. In patients with multiple previous abdominal operations and recurrent bowel obstruction episodes, SBO is probably more likely to resolve with conservative treatment.

Author contributions

L.J, C.N. and U.L. contributed to the conception and design. L.J. collected the data and wrote the article. Ü.K. was responsible for statistical analysis. C.N. and U.L. performed the critical revision of the article.

Disclosure

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References

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